

Conservation Program design, Enrollment and Cost-Effectiveness in Colorado Republican River Basin

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Abstract

This paper examines participation behavior related to irrigated land conservation and assesses the cost-effectiveness of using a single program versus a mix of programs to meet a given acreage-retirement target. We develop a theoretical model in which heterogeneous producers self-select into programs based on their opportunity cost of irrigation, showing that the relative cost-effectiveness of a single-program versus a multi-program design depends on the targeted acreage and the share of low-opportunity-cost producers available. We then estimate a multinomial logit model of program enrollment using field-level data for Colorado's Republican River Basin. Consistent with the theoretical prediction, the results show that producers' choice between the conservation programs is influenced by factors that determine the opportunity cost of irrigation, such as land capability class and climate. These findings provide empirical evidence on the drivers of participation in conservation programs and inform policy design to achieve groundwater conservation targets at the lowest cost in the Republican River Basin.

1. Introduction

In the United States, as in many parts of the world, agricultural policy has increasingly relied on voluntary instruments to achieve environmental and land-use objectives. These policies generally provide financial incentives to private landowners or firms in exchange for voluntary changes in land use or management practices that improve environmental outcomes. In the United States, the Conservation Reserve Program (CRP) was established under the 1985 Farm Bill to protect environmentally sensitive cropland by encouraging landowners to retire highly erodible land from agricultural production and enroll it in conservation cover. Administered as a voluntary program by the United States Department of Agriculture (USDA), CRP offers landowners and operators contracts with fixed terms of 10 to 15 years, during which enrolled land is removed from crop production. (Caldas et al., [2016](#))

In 1996, two separate programs were established to motivate agricultural land conservation, and both are currently being used to conserve groundwater by taking irrigated land out of production. The Environmental Quality Incentives Program (EQIP) was established under the Federal Agriculture Improvement and Reform Act of 1996 as a key USDA initiative to promote conservation on working lands, and it has since been reauthorized and refined in subsequent Farm Bills. EQIP funds several subprograms, such as the Agricultural Water Enhancement Program (AWEP) and the Conservation Innovation Grants program (Stubbs, [2010](#)). It has been used specifically to retire irrigated acreage for groundwater conservation beyond Colorado. In Nebraska, AWEP funded the Upper and Lower Republican Natural Resources District projects as well as the Ogallala/Platte River Recovery project to convert irrigated cropland to non-irrigated use along the Platte River (Natural Resources Conservation Service, [2011](#)).

The Conservation Reserve Enhancement Program (CREP) was also established the same year as a supplementary component of the CRP. It operates as a federal–state partnership, administered by the Farm Service Agency, that provides incentives to landowners who enroll eligible land in conservation practices. It is administered through agreements

tailored to each state or region, so its conservation objectives vary considerably. In many western states, including Colorado, Kansas, and Idaho, CREP is used primarily to reduce groundwater use on irrigated land (Idaho Soil and Water Conservation Commission, [n.d.](#)). In contrast, in other regions, such as Minnesota, the program emphasizes water quality, wetland restoration, and wildlife habitat (U.S. Department of Agriculture, Farm Service Agency, [2025](#)).

Both programs reduce consumptive groundwater use primarily by removing irrigated acreage from active production. When enrolled land is converted from irrigated cropland to non-irrigated crops or permanent cover (such as native grasses), the irrigation water demand that previously required groundwater pumping is eliminated, directly reducing withdrawals from the aquifer (USDA, [2023](#)). In Colorado, the Republican River Water Conservation District (RRWCD) is using both CREP and EQIP to retire irrigated land and comply with the Republican River Compact Agreement.

However, enrollment in these programs is shaped by heterogeneous farm, soil, and aquifer characteristics, and different conservation programs impose distinct restrictions and offer different incentive levels (Adjei et al., [2024](#); Thapa et al., [2024](#)). As a result, relying on a single program may raise the cost of achieving a given land-retirement target or vice versa. Combining programs may improve cost-effectiveness by allowing producers to self-select into contracts that better match their opportunity costs. Despite the widespread use of multiple conservation programs, there is limited empirical evidence on how alternative program mixes affect the total cost of meeting acreage-based conservation goals and the factors that drive producers' enrollment choices.

This gap is not only academic. In Colorado's Republican River Basin, compliance with a decades-old interstate water compact now hinges on retiring a specific acreage target by a fixed deadline, making the cost-effectiveness of program design a critical contemporaneous policy question as well. This paper addresses two related questions: What factors influence producers' decisions to enroll irrigated land in CREP versus EQIP? And does a single program or a combination of programs deliver the lowest-cost way to meet a given acreage-retirement target?

This paper makes two main contributions. First, we develop a theoretical framework that formalizes the conditions under which a single conservation program versus a mix of programs achieves a given acreage retirement target at the lowest cost. Second, by jointly modeling enrollment across two coexisting programs rather than treating enrollment as a single binary decision, our approach captures how producers self-select across programs based on differing opportunity costs, contract restrictions, and incentive levels, offering more direct empirical evidence on program self-selection than prior studies. Together, the theoretical and empirical results offer practical guidance for policymakers in the Republican River Basin, and more broadly for conservation programs elsewhere, on how to design and combine incentive programs to meet land retirement targets at the lowest cost.

The theoretical model shows that the least-cost program design depends on the size of the acreage-retirement target relative to the supply of low-cost producers. A single uniform incentive program is least-cost when the target can be met entirely by low-cost types, while a two-program design becomes more cost-effective as the target grows to require enrollment of higher-cost producers. Our empirical analysis shows that producers self-select across CREP and EQIP based on their underlying opportunity costs. Fields with land poorly suited to non-irrigated production, and therefore a higher opportunity cost of irrigation, are more likely to enroll in CREP's stricter, higher-paying contracts, while fields better suited to non-irrigated production are more likely to enroll in EQIP's less restrictive, lower-paying program. Enrollment in both programs also responds to other factors, such as financial incentives and climatic and aquifer conditions, that shape the relative returns to continued irrigation.

This study is organized as follows. Section 1 provides an introduction and background on land retirement programs in the United States, with particular emphasis on the Republican River Basin. Section 2 reviews the relevant literature on land retirement, conservation programs, and farmer enrollment decisions. Section 3 develops a theoretical framework that describes how farmers choose among alternative conservation programs and analyzes the cost implications of using one versus multiple programs to achieve a given acreage retirement target. Section 4 presents the empirical model and data used

in the analysis, drawing on well-level and spatial information from the Republican River Basin. Section 5 discusses the empirical results and their implications. Finally, Section 6 concludes with a summary of key findings, policy recommendations, and a discussion of the study’s limitations.

2. Background

The Republican River Basin is located in northeast Colorado and extends into southwest Nebraska and northwest Kansas. The Colorado portion of the basin fully covers Phillips and Yuma counties and partially extends into Logan, Washington, Sedgwick, Kit Carson, and Lincoln counties. The surface water sources in the basin are the North and South Forks of the Republican River, the Arikaree River, and Beaver Creek. Groundwater in the basin is sourced primarily from the Ogallala Aquifer. Figure 1 shows the map of the part of the Republican River Basin, the South Fork Focus Zone¹, and the location of wells enrolled in CREP and EQIP.

The basin is shared among Kansas, Nebraska, and Colorado, and surface water is governed by the Republican River Compact of 1942. However, widespread groundwater use in Colorado’s portion of the basin has led to declining groundwater stocks and reduced surface stream flows. Over the years, the three states have had disputes over compliance with the compact. As part of the most recent settlement, Colorado agreed to retire an additional 25,000 acres of irrigated land in the South Fork Focus Zone of the basin by 2030 (Loos et al., 2022). The RRWCD in Colorado is therefore using CREP and EQIP to retire irrigated land. It offers incentives for landowners to remove their land from irrigated production and relinquish their associated water rights. The conservation programs are expected to increase surface water streamflow, thereby bringing the state into compliance with the compact.

In Colorado, CREP began in 2006 and is part of the larger CRP program. Through CREP, federal and state resources are made available to producers to voluntarily enroll

¹The South Fork Focus Zone is the region along the South Fork of the Republican River where Colorado must retire 25,000 acres of irrigated land by 2030 to stay in compliance with the Republican River Compact.

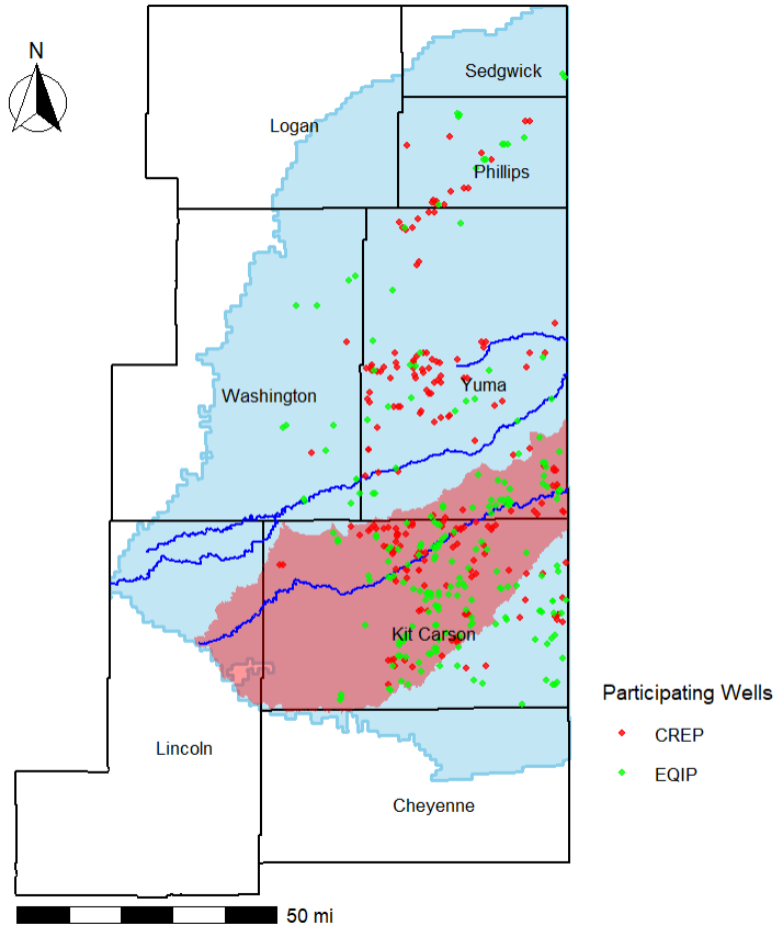


Figure 1: Map of Study Area and Retired Wells

in 14- to 15-year contracts. Participants remove cropland from irrigated agricultural production and convert it to grasses, trees, or other approved vegetation. Irrigation on enrolled acres is not allowed except when establishing permanent cover. Rain-fed crop production is also not allowed during the contract period. Producers permanently give up their water rights and plug the associated wells. The goals of this program in the Republican River basin are to reduce irrigation water use for agriculture from the Ogallala Aquifer by 6.5 percent from 2004 levels, improve water quality through wetland restoration, reduce sediment, nutrients, and other pollutants entering water bodies, and improve stream flows, among others (USDA, 2023).

On the other hand, EQIP is implemented by the Natural Resources Conservation Service and offers producers, ranchers, and private forest landowners cost-share assistance to implement structural and management conservation practices that provide environmental benefits on working agricultural lands. Producers sign a contract to receive an

annual payment for implementing approved conservation programs and abandoning the well. According to the Natural Resources Conservation Service, the contract period is 5 years and may be extended to no more than 10 years (NRCS, 2022). During this contract period, landowners may engage in non-irrigated crop production as part of an approved conservation program. The landowner has one year from the date of contract signing to permanently shut down the well.

Producers enrolled in conservation programs receive financial incentives from the USDA and RRWCD. The payment rates are determined by factors such as county rental rates, location relative to the South Fork Focus Zone of the Republican River, land type, and the conservation practice implemented (USDA-FSA, 2025). Table 1 shows the RRWCD rates per acre. CREP participants within the South Fork Focus Zone region received \$2,155 in 2022, \$2,780 in 2023, and \$2,690 in 2024, while rates outside the zone remained fixed at \$90 per acre. EQIP payments within the focus zone were \$3,761, \$3740, and \$3651 in the respective years, whereas payments outside the region were constant at \$240 per acre. As of August 2025, a total of 62,396 acres have been enrolled in the basin, 31,363 acres through CREP and 31,034 acres through EQIP.

Table 1: RRWCD CREP and EQIP Incentive Rates per acre

Year	South Fork		Rest of the Basin	
	Max/Acre	Min/Acre	Max/Acre	Min/Acre
CREP Rates				
2024	\$2,690	\$2,690	\$90	\$90
2023	\$2,780	\$2,780	\$90	\$90
2022	\$2,155	\$2,120	\$90	\$90
EQIP Rates				
2024	\$3,651	\$2,984	\$240	\$240
2023	\$3,740	\$3,651	\$240	\$240
2022	\$3,761	\$3,011	\$240	\$240

Table 2 shows the USDA payment rates across counties in the Republican River Basin. Between 2022 and 2025, CREP incentive rates varied noticeably across the basin. CREP payments are based on the county average soil rental rate and soil productivity (USDA-FSA, 2025). Yuma County consistently received the highest payments, rising from \$222 per acre in 2022 to a peak of \$257 per acre in 2024, before slightly declining to \$239 in

2025. In contrast, Lincoln County had among the lowest rates in most years (lowest in 2022, tied for lowest in 2023 and 2025), though its rate rose above those of several other counties in 2024. Counties such as Phillips and Logan saw relatively higher payments compared to the basin average, while Cheyenne, Kit Carson, and Washington remained in the mid-range. EQIP payments, by contrast, are uniform across the basin and have gradually increased from \$142 per acre in 2022 to \$215 per acre in 2025.

Table 2: USDA Incentive Rates

County	2022 Rate	2023 Rate	2024 Rate	2025 Rate
CREP Rates				
Cheyenne	\$178.00	\$164.00	\$184.00	\$187.00
Kit Carson	\$178.00	\$171.00	\$184.00	\$190.00
Lincoln	\$132.00	\$164.00	\$200.00	\$171.00
Logan	\$183.00	\$194.00	\$230.00	\$201.00
Phillips	\$195.00	\$237.00	\$240.00	\$221.00
Sedgwick	\$193.00	\$164.00	\$188.00	\$171.00
Washington	\$193.00	\$184.00	\$217.00	\$181.00
Yuma	\$222.00	\$251.00	\$257.00	\$239.00
EQIP Rates				
Basin-Wide	\$142.01	\$159.77	\$186.90	\$215.28

3. Literature Review

The literature on agricultural land retirement and conservation programs relevant to this study can be grouped into several broad strands. One strand examines the farm and county-level factors that drive landowner enrollment decisions. Another strand considers the environmental and economic impacts of CRP. Another looks at the costs that the retirement of irrigated land imposes on communities. Notably, most of these studies treated CRP as a single, undifferentiated program, without disaggregating participation or impacts across its distinct programs, such as CREP and EQIP.

3.1 Participation in Conservation Programs

A number of studies examine the factors that drive producers to participate in conservation programs. These studies point to a fairly consistent set of drivers: enrollment tends to rise with payment levels, erosion risk, land size, and access to information, while it falls

with land value, urban influence, and farming being the operator’s primary occupation. Several studies also highlight that awareness of program options and demographic factors, such as gender and education, shape participation in ways not fully captured by financial incentives alone. The remainder of this review turns to the broader literature on program participation through land-use changes and enrollment behavior.

Jones (2007), Lubell et al. (2013) and Monger et al. (2018) illustrate these patterns in more detail. Jones (2007) found that female producers were more likely to participate in EQIP, and that participation was positively associated with erosion levels, the presence of pastureland, payment per acre, and production costs, but negatively associated with farm value and with farming being the operator’s primary occupation. Lubell et al. (2013) similarly found that larger land size, greater access to information sources, and having a succession plan in place all increased participation, while off-ranch income and higher education levels specifically raised EQIP participation. Notably, producer income was not a significant predictor of participation in their sample.

Monger et al. (2018) focused on the financial incentives offered to producers to enroll in CREP. Their results show that enrollment is predicted to increase by 0.087 percentage points with a \$10 increase in the incentives offered. Similar to Jones (2007) and Lubell et al. (2013), they found that the probability of enrollment is also influenced by the aquifer’s saturated thickness and the soil characteristics that impact land productivity. Producers were more likely to enroll when payments were higher and when the underlying aquifer had less water available. However, the study is limited to CREP as a single program, so it does not address how producers choose between CREP and EQIP when both are available.

Thapa et al. (2024) highlighted a factor largely overlooked in the studies above: landowners’ awareness of CRP’s benefits in improving water quality and availability was among the strongest predictors of enrollment intention, alongside social influence from neighboring landowners. This underscores that information gaps, not just financial or attitudinal barriers, may constrain CRP participation.

Beyond these individual and informational drivers, participation also responds to the broader market environment in which enrollment decisions are made. Adjei et al. (2024) examined how CRP rental rates influence enrollment under different crop price regimes, allowing the responsiveness of enrollment to rental rate incentives to vary with crop prices. They found a nonlinear relationship between rental rates and enrollment: a 10 percent increase in CRP rental rates raised enrollment by 5.1 percent at baseline crop prices, but this effect fell to 4.2 percent once crop prices crossed a specific threshold.

Their simulations further showed that higher crop prices reduce enrolled acreage and its associated environmental benefits, lowering carbon sequestration while increasing phosphorus and nitrogen runoff and sediment loss. This finding reinforces a point relevant to program cost: the payment needed to induce a given level of enrollment is not fixed but rises with the opportunity cost of retiring land from production, meaning the cost of reaching a retirement target will vary based on market-based factors.

Chaudhary et al. (2026) noted that many of these studies relied on cross-sectional survey data to examine CRP participation at a single point in time. This approach implicitly treats currently unenrolled producers as permanent non-participants. However, CRP enrollment is typically a complex, multi-stage process (Hellerstein, 2017). Chaudhary et al. (2026) argued that it is essential to consider enrollment patterns over time.

To address this gap, they employed a discrete-time duration analysis to examine the factors influencing the time elapsed before producers enrolled in General CRP, measured from either the initiation of CRP in 1985 or the year the producer began managing/operating a farm, whichever was later. Their model assumed that producers not currently enrolled could still enroll in the future under favorable conditions. Their findings showed that younger producers took less time to enroll than older producers, and that larger total landholdings, greater awareness of CRP, and positive perspectives on the enrollment ranking process were all associated with faster enrollment.

Beyond identifying the factors that influence enrollment decisions, some studies have sought to measure the actual level of CRP participation by tracking observable changes

in land use over time. Yuan et al. (2015) used multi-temporal remote sensing imagery from 1985, 1995, 2005, and 2013 to track land-use change in the Minnesota River Valley across the transition from having no long-term conservation program to an era of active CRP enrollment. They found that about 89,000 acres of cropland were converted to grassland or forest over this 28-year period, and that cropland area continued to decline even as rising corn and soybean prices would have been expected to pull land back into production.

Harp and Maczko (2024) surveyed producers in the Upper Great Plains whose CRP contracts had expired within the prior seven years, comparing those who returned to crop production with those who instead enrolled in EQIP after CRP; producers who transitioned out of CRP tended to return to crop production, while EQIP enrollees shifted more toward cattle and grazing, and only 58 percent of the CRP group reported being aware that EQIP was an option, pointing to a program-awareness gap that likely affects enrollment patterns independent of producer preferences.

Post-contract decisions offer a further lens on CRP participation. Harp and Maczko (2024) surveyed producers in the Upper Great Plains whose CRP contracts had expired within the prior seven years, comparing those who returned to crop production with those who instead enrolled in EQIP after CRP. Producers who transitioned out of CRP tended to return to crop production, while EQIP enrollees shifted more toward cattle and grazing. Only 58 percent of the CRP group reported being aware that EQIP was an option, pointing to a program-awareness gap that likely affects enrollment patterns independent of producer preferences.

3.2 Cost of Conservation Programs

We begin with a set of studies that most directly analyze the costs of conservation programs (Cramton et al., 2021; Johnson et al., 2016; Miao et al., 2016). These studies approach cost from opposite ends of the program: one values the ecosystem service benefits generated per dollar of rental payment, and the other examines how the auction mechanism that sets those payments shapes the rents captured by participating farmers.

Together, they offer the most direct cost evidence available, but neither addresses the question central to this study: the total cost of using a single program or a combination of programs to reach a given land-retirement target.

Johnson et al. (2016) conducted a benefit-cost analysis of CRP land retirement in the Indian Creek watershed in Iowa, quantifying the value of ecosystem services, including flood damage reduction, water and air quality improvements, and greenhouse gas mitigation, generated by enrolled land relative to the rental payments farmers received. Across all scenarios tested, ecosystem service benefits exceeded program costs: expanding CRP enrollment would require an average per-acre payment of \$1,311 over a 10-year contract, but generate benefits with a net present value of \$1,710 to \$6,401. This finding suggests CRP investment is justified on a benefit-cost basis in this watershed.

Cramton et al. (2021) used laboratory experiments to evaluate how CRP's General Signup and auction design affect program cost-effectiveness, focusing on the price-cap mechanism the program uses to constrain rents paid to enrolled farmers. They found that tightening bid caps reduced participation, even among low-cost producers, undermining cost-effectiveness, while loosening caps too far allowed sellers to extract higher rents without meaningfully increasing competition. They further concluded that a meaningful share of CRP's cost per acre retired is attributable not only to the underlying opportunity cost of land but also to auction design choices that shape how much rent producers can capture.

Another set of studies, such as Harp and Maczko (2024), Yuan et al. (2015) and Zhang et al. (2025), approaches cost analysis indirectly via proxies such as yield and land-use changes, rather than through direct estimates of the costs these programs impose on communities or taxpayers.

Zhang et al. (2025) found that a 10 percent increase in EQIP payments raised corn yields by about 0.014 bushels per acre in the following year and by up to 0.085 bushels per acre when payments were sustained over five years; a 10 percent increase in CRP payments over five years raised soybean yields by at least 0.040 bushels per acre. This

implies that conservation spending generates a private productivity return to enrolled farms alongside its intended environmental benefits.

Beyond this, the literature does not address the implications of the multi-program design we observe in practice, where CRP, EQIP, and other conservation programs co-exist and compete for the same producers. In particular, it doesn't provide evidence on the total cost of using a single program versus a combination of programs to reach a given land-retirement target. This study addresses that gap in two ways. We develop a theoretical framework that outlines the conditions under which a single conservation program, versus a mix of programs, achieves a given acreage-retirement target at the lowest cost. Second, we model enrollment across two conservation programs, CREP and EQIP, capturing how producers self-select based on differing opportunity costs, contract restrictions, and incentive levels.

4. Theoretical Model

In this section, we explore the conditions under which multiple programs achieve conservation objectives at the lowest total cost to policymakers compared with a single program. Classical land-use theory, originating in the nineteenth century, posits that land-use decisions are made to maximize quasi-rents, defined as revenues net of variable costs (Roberts & Lubowski, 2007). Under this framework, both land rents and land uses vary systematically with land characteristics, such as soil fertility, climate, location, and other physical attributes, that influence the relative profitability of alternative activities. A substantial empirical literature supports these core predictions, showing that quasi-rent maximization governs land-use choices and that land characteristics shape both land rents and observed land-use patterns. (e.g., Lichtenberg 1989; Stavins and Jaffe 1990; Plantinga 1996 cited in Roberts and Lubowski (2007)). Our theoretical model follows this approach. We assume that producers make conservation decisions to maximize their utility.

We compare the programmatic cost of a single conservation program with that of two programs and examine the conditions under which total costs are lower for each approach.

Using two programs is only feasible when a separating equilibrium exists, so that utility-maximizing producers self-select into the program best suited to their type. In this setting, the planner designs a menu of incentive payments and associated requirements that induce sorting across programs, satisfying both incentive compatibility and individual rationality.

Consider two alternative irrigated land conservation programs available to the planner. Program A allows continued use of the land under non-irrigated production after retirement, whereas Program B prohibits any form of agricultural production. Assume θ is an index of the farm type determined by its sensitivity to climate, soil type, and other environmental factors. High values of θ mean the land is more vulnerable to climate and produces lower non-irrigated profits, and thus has a high opportunity cost of irrigation. We can denote the irrigated production profit for the individual i as $\Pi_i^R(\theta)$. The non-irrigated profit for land enrolled in A is $\Pi_i^{NR}(\theta)$.

A producer will choose to participate in Program A if the payoff, incentive plus the non-irrigated profit, exceeds both the profit from continued irrigated production and the profit from enrolling in Program B. Similarly, a producer will enroll in Program B if its payoff exceeds the profit from continued irrigation and from enrolling in Program A. The planner can design a single conservation program to enroll all types, or two conservation programs for producers to self-select based on their type, θ . With two programs, the objective would be to design the programs so that producers with low opportunity cost (low cost types) enroll in program A, which offers low incentives and allows non-irrigated production, while those with high opportunity cost (high cost types) enroll in program B, where they can get higher incentives and do not engage in non-irrigated production.

To establish the existence of an incentive pair (I_A, I_B) that induces self-sorting into a separating equilibrium, we model this as an adverse selection problem.

4.1 Model Setup

4.1.1 Assumptions

- i. We assume producers are one of two types, $\theta \in \{\theta_L, \theta_H\}$, where θ_L denotes a producers with low θ and θ_H a high θ producer type. We can define the opportunity cost of irrigation ($OC(\theta_i)$) as the difference between the profit from irrigation and the profit from non-irrigation.

$$OC(\theta_i) = \Pi^R(\theta_i) - \Pi^{NR}(\theta_i) \quad (1)$$

- ii. Both irrigated and non-irrigated profits decline with θ . However, irrigated profit is weakly higher for θ_L , and non-irrigated profit is strictly higher for low cost types θ_L

$$\Pi_i^R(\theta_L) \geq \Pi_i^R(\theta_H), \quad (2)$$

$$\Pi_i^{NR}(\theta_L) > \Pi_i^{NR}(\theta_H)$$

Non-irrigated profits are more sensitive to changes in θ than irrigated profits. Therefore, the gap in non-irrigated profits across types is larger than the gap in irrigated profits across types. In other words, the opportunity cost of giving up non-irrigated farming to take a full retirement program is lower for type θ_L .

$$\pi^R(\theta_L) - \pi^R(\theta_H) < \pi^{NR}(\theta_L) - \pi^{NR}(\theta_H) \quad (3)$$

4.1.2 Producer Payoffs

The planner offers two retirement contracts:

- i. Program A offers an incentive of I_A and allows non-irrigated production. The utility of enrolling in A is:

$$U_A(\theta) = I_A + \pi^{NR}(\theta) \quad (4)$$

- ii. Program B offers an incentive I_B and prohibits non-irrigated production. The utility

of enrolling in B is:

$$U_B(\theta) = I_B \quad (5)$$

iii. Outside Option (Continue Irrigation) has a payoff of $\pi^R(\theta)$ and a utility of

$$U_0(\theta) = \pi^R(\theta) \quad (6)$$

4.1.3 Conditions for a Separating Equilibrium

Because type θ_L generates relatively higher returns on non-irrigated farming compared to θ_H , economic efficiency requires type θ_L to select Program A, which pays only for the opportunity cost of irrigation and allows non-irrigated production, and type θ_H selects Program B. For a pair of incentives (I_A, I_B) to induce separation, four constraints must hold simultaneously:

i. Incentive Compatibility for θ_L (IC_L): Producer of type θ_L will choose program if

$$\begin{aligned} I_A + \pi^{NR}(\theta_L) &\geq I_B \\ \Rightarrow I_B - I_A &\leq \pi^{NR}(\theta_L) \end{aligned} \quad (7)$$

ii. Incentive Compatibility for θ_H (IC_H):

$$\begin{aligned} I_B &\geq I_A + \pi^{NR}(\theta_H) \\ \Rightarrow I_B - I_A &\geq \pi^{NR}(\theta_H) \end{aligned} \quad (8)$$

iii. Individual Rationality for θ_L (IR_L):

$$\begin{aligned} I_A + \pi^{NR}(\theta_L) &\geq \pi^R(\theta_L) \\ \Rightarrow I_A &\geq \pi^R(\theta_L) - \pi^{NR}(\theta_L) \end{aligned} \quad (9)$$

iv. Individual Rationality for θ_H (IR_H):

$$I_B \geq \pi^R(\theta_H) \quad (10)$$

The combination of IC_L and IR_L ensures that a producer of type θ_L prefers program A to both program B and remaining in production. Specifically, IC_L ensures that program A provides at least as much utility as program B, while IR_L ensures that participation in program A provides at least as much utility as remaining in production. Thus, when both conditions hold, type θ_L retires and selects program A. Similarly, the combination of IC_H and IR_H ensures that a producer of type θ_H prefers program B to both program A and remaining in production. IC_H makes program B preferable to program A, while IR_H ensures that participation in program B is preferable to remaining in production. Hence, type θ_H retires and selects program B.

4.2 Proof of Separating Equilibrium

To establish the existence of a separating equilibrium, we show that the planner can design a pair of incentive payments (I_A^*, I_B^*) such that low-opportunity-cost producers (θ_L) choose Program A, high-opportunity-cost producers (θ_H) choose Program B, and neither type chooses to mimic the other type.

4.2.1 Step 1: Feasibility of Incentive Compatibility

Combining (IC_L) and (IC_H) defines the feasible spread between the two incentive payments:

$$\pi^{NR}(\theta_H) \leq I_B - I_A \leq \pi^{NR}(\theta_L) \quad (11)$$

By Assumption (ii), non-irrigated profit strictly declines in type ($\pi^{NR}(\theta_L) > \pi^{NR}(\theta_H)$). Consequently, the interval $[\pi^{NR}(\theta_H), \pi^{NR}(\theta_L)]$ is non-empty and has strictly positive measure. Any difference $\Delta I \equiv I_B - I_A$ within this interval satisfies both incentive compatibility constraints simultaneously.

4.2.2 Step 2: Constructing the Incentive Structure

In a cost-minimizing separating equilibrium, the planner sets payments so that the individual rationality constraint for the high-opportunity-cost type binds, and the incentive compatibility constraint for the high type binds, preventing them from taking Program A.

- When IR_H binds, I_B^* is set to the minimum payment that induces type θ_H to participate:

$$I_B^* = \pi^R(\theta_H) \quad (12)$$

- Binding IC_H and substituting I_B^* gives I_A^* and sets the feasible spread to the lower bound $\Delta I = \pi^{NR}(\theta_H)$, which minimizes $I_B - I_A$:

$$\begin{aligned} I_B - I_A &= \pi^{NR}(\theta_H) \\ \implies I_A &= I_B - \pi^{NR}(\theta_H) \\ \implies I_A^* &= \pi^R(\theta_H) - \pi^{NR}(\theta_H) \end{aligned} \quad (13)$$

4.2.3 Step 3: Verification of All Constraints

We verify that the proposed general incentive structure $(I_A^*, I_B^*) = (\pi^R(\theta_H) - \pi^{NR}(\theta_H), \pi^R(\theta_H))$ satisfies the four constraints:

1. Individual Rationality of θ_H (IR_H):

$$I_B^* = \pi^R(\theta_H) \geq \pi^R(\theta_H) \quad (14)$$

Holds with equality.

2. Incentive Compatibility of θ_H (IC_H):

$$\begin{aligned} I_B^* - I_A^* &= \pi^R(\theta_H) - [\pi^R(\theta_H) - \pi^{NR}(\theta_H)] \\ \implies I_B^* - I_A^* &= \pi^{NR}(\theta_H) \end{aligned} \quad (15)$$

Therefore, IC_H holds with equality (broken in favor of Program B by an arbitrarily small $\epsilon > 0$).

3. Incentive Compatibility of θ_L (IC_L):

$$I_B^* - I_A^* = \pi^{NR}(\theta_H) < \pi^{NR}(\theta_L) \quad (16)$$

Holds strictly because $\pi^{NR}(\theta_L) > \pi^{NR}(\theta_H)$. Type θ_L strictly prefers Program A over Program B.

4. Individual Rationality of θ_L (IR_L):

Type θ_L accepts Program A if $I_A^* + \pi^{NR}(\theta_L) \geq \pi^R(\theta_L)$. Substituting I_A^* :

$$\pi^R(\theta_H) - \pi^{NR}(\theta_H) + \pi^{NR}(\theta_L) \geq \pi^R(\theta_L) \quad (17)$$

Rearranging terms yields:

$$\pi^{NR}(\theta_L) - \pi^{NR}(\theta_H) \geq \pi^R(\theta_L) - \pi^R(\theta_H) \quad (18)$$

This holds strictly under Assumption (ii). Thus, type θ_L strictly prefers participating in Program A over continuing irrigation.

4.2.4 Conclusion

A non-empty set of incentive pairs (I_A, I_B) exists such that:

$$\begin{aligned} I_B &= \pi^R(\theta_H), \\ I_A &\in [\pi^R(\theta_L) - \pi^{NR}(\theta_L), \pi^R(\theta_H) - \pi^{NR}(\theta_H)] \end{aligned} \quad (19)$$

Under any pair in this set:

- Producers of low cost types (θ_L) choose Program A.
- Producers of high cost types (θ_H) choose Program B.
- Neither type has an incentive to mimic the other or remain fully irrigated.

The intuition is that I_A needs only to cover the marginal opportunity cost of forgoing irrigation, since the producer retains non-irrigated profit $\Pi^{NR}(\theta_i)$ under Program A. Program B, by contrast, doesn't have non-irrigated production profit, so I_B must cover the full value of irrigated production. This asymmetry generates sorting: low-cost types, with lower opportunity cost, are cheaply retained by A's modest incentive, while high-cost types require B's higher payment. Figure 2 shows the incentive structure.

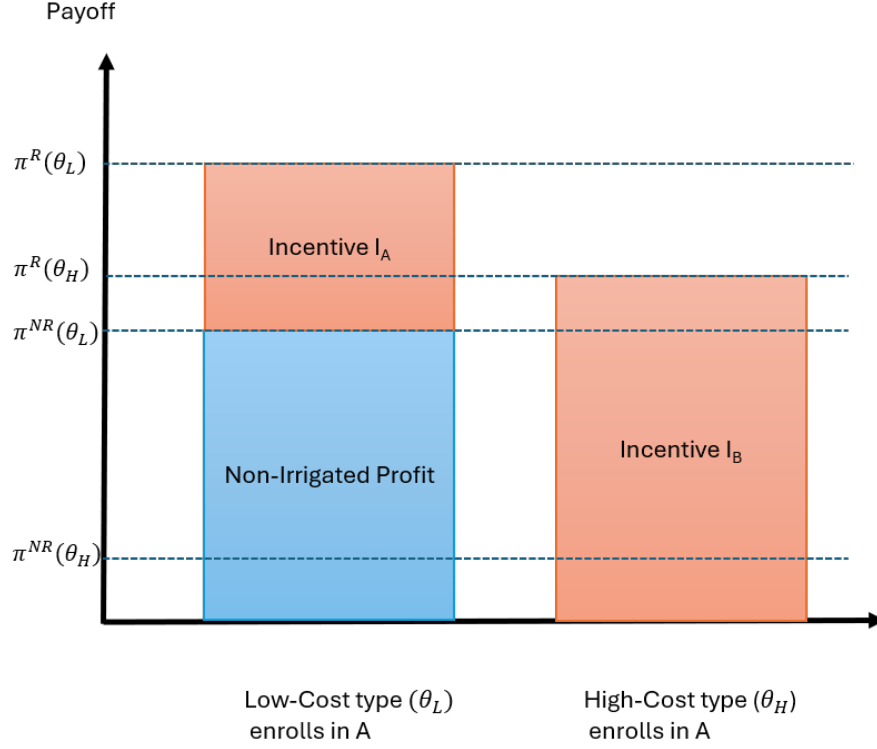


Figure 2: Incentive Structure

4.3 Cost Functions

4.3.1 One Conservation Program

Consider the case where there is one conservation program to enroll in, Program A, which offers a uniform incentive rate of I_0 and allows non-irrigated farming activities. The program pays only for the opportunity cost of irrigation. A risk-neutral, profit-maximizing producer will enroll if:

$$I_0 \geq \Pi^R(\theta_i) - \Pi^{NR}(\theta_i) \quad (20)$$

By adjusting the level of incentives offered, policymakers can influence the number of acres retired from irrigation. A lower incentive rate attracts low-cost types, while a higher incentive rate attracts both low- and high-cost types. This relationship allows policymakers to control the total acreage enrolled by adjusting the incentive. Policy cost of the retiring η acres is:

$$TC_1 = \eta * I_0 \quad (21)$$

4.3.2 Two Conservation Program

Now, assume there are two conservation programs, A and B, which offer different incentive rates. The incentive rate for A is denoted by I_A , while that for B is denoted by I_B , where $I_A < I_B$. Program A targets low-cost types (θ_{iL}) and allows non-irrigated farming, while program B targets high-cost types (θ_{iH}) and does not allow non-irrigated farming after retirement. The low-cost type producer will enroll in A if:

$$I_A \geq \Pi^R(\theta_{iL}) - \Pi^{NR}(\theta_{iL}) \quad (22)$$

A high-cost producer will enroll in B if:

$$I_B \geq \Pi^R(\theta_{iH}) \quad (23)$$

Let η be the number of acres to be retired and β be the proportion of acres that are low-cost. The cost minimizing policymaker pays the minimum incentive possible, $I_A = \Pi^R(\theta_{iL}) - \Pi^{NR}(\theta_{iL})$ and $I_B = \Pi^R(\theta_{iH})$. We can assume the same relationship between acres enrolled in each program and incentives as:

$$\beta\eta = f(I_A) \quad \text{and} \quad (1 - \beta)\eta = f(I_B) \quad (24)$$

Thus, the total cost of the program is:

$$TC_2 = \beta\eta I_A + (1 - \beta)\eta I_B \quad (25)$$

To assess which program design as a lower policy cost for meeting a given acreage-retirement target, we compare the two cost functions, $TC_1 = \eta I_0$ and $TC_2 = \beta\eta I_A + (1 - \beta)\eta I_B$. However, this comparison is only meaningful under the separating-equilibrium or self selecting conditions described earlier, which ensure that low-cost and high-cost fields self-select into their respective programs. Only when a separating equilibrium holds do the expressions I_A and I_B correctly reflect the incentive structure, allowing us to determine which design yields lower total cost.

4.4 Cost Comparison

We compare the cost functions $TC_1 = \eta * I_0$ and $TC_2 = \beta\eta I_A + (1 - \beta)\eta I_B$. The minimum incentive producers can accept are:

$$\begin{aligned} I_A &= \Pi^R(\theta_{iL}) - \Pi^{NR}(\theta_{iL}), \\ I_0 &= \Pi^R(\theta_i) - \Pi^{NR}(\theta_i), \\ I_B &= \Pi^R(\theta_i) \end{aligned} \tag{26}$$

If the retirement goal is small and can be filled by low-cost types only, then $I_0 = \Pi^R(\theta_{iL}) - \Pi^{NR}(\theta_{iL})$ and thus $I_A = I_0$. If the retirement goal is high and must be filled by both low and high cost types, then $I_0 = \Pi^R(\theta_{iH}) - \Pi^{NR}(\theta_{iH})$, thus $I_A < I_0$. Therefore $I_A \leq I_0$. We also know that $I_0 < I_B$ because I_0 is the rate that high-cost producers accept when allowed to earn additional income from non-irrigated production, while I_B is the minimum rate they would accept without non-irrigated production. Therefore, to enroll both low and high cost types, the incentive levels are

$$I_A \leq I_0 < I_B \tag{27}$$

The cost-effectiveness of using one or two programs depends on the proportion of acres to be retired and the available low-cost types. To analyze cost-effectiveness, we consider two cases: targeting a small proportion of acres that can be filled with available low-cost types, and targeting a large proportion of acres that cannot be filled with available low-cost types.

4.4.1 Case One: Low target acres

Consider the case where authorities aim to retire a small proportion η^* such that η^* falls entirely within the supply of low-cost types β (i.e., $\eta^* \leq \beta$). The incentives are:

$$\begin{aligned} I_A &= \Pi^R(\theta_{iL}) - \Pi^{NR}(\theta_{iL}), \\ I_0 &= \Pi^R(\theta_{iL}) - \Pi^{NR}(\theta_{iL}) \end{aligned} \tag{28}$$

Thus

$$I_A = I_0 < I_B \quad (29)$$

A single program A that pays the rate $I_A = I_0$ attracts all low-cost producers. If a second program, B, paying I_B , were instead offered to enroll some high-cost producers, with the policymaker allowing the target to be filled by a mix of producer types, the total program cost would be higher than under a single program. Figure 3 shows the incentive rate. The cost of retiring land is the area under the incentive level. The left panel shows the cost when all acres are retired using a single program with an incentive rate of I_0 . The right panel shows the cost region when some acres are retired at an incentive rate I_A and others at I_B . In this scenario, the least-cost option is to use a single program that covers only the opportunity cost of irrigation.

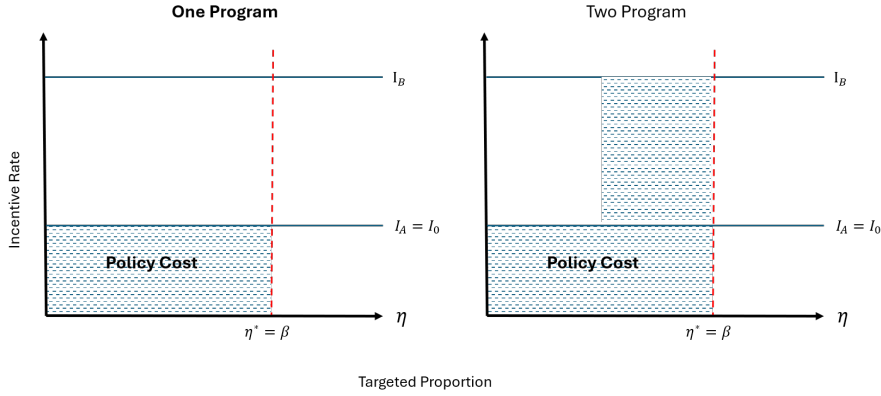


Figure 3: Case one, cost of retirement for low target acres

4.4.2 Case Two: High target acres

Now consider the case in which the authorities aim to retire a proportion of acres that exceeds the amount that can be covered by the proportion of available low-cost land types ($\eta^* > \beta$). In this situation, even if all low-cost acres are enrolled, the policymaker must still draw in some higher-cost land to meet the retirement goal. The single-program design must set the incentive high enough to attract both low- and high-cost types.

$$I_0 = \Pi^R(\theta_{iH}) - \Pi^{NR}(\theta_{iH}) \quad (30)$$

Thus

$$I_A < I_0 < I_B \quad (31)$$

The cost effectiveness depends on the relative levels of both β and η^* . Figure 4 shows the comparisons for all four scenarios. When the proportion of low-cost types, β is low, using a single program that pays for the opportunity cost of irrigation I_0 is lower cost than two programs. The opposite is true; it is cost-effective to use two programs as β approaches η^* .

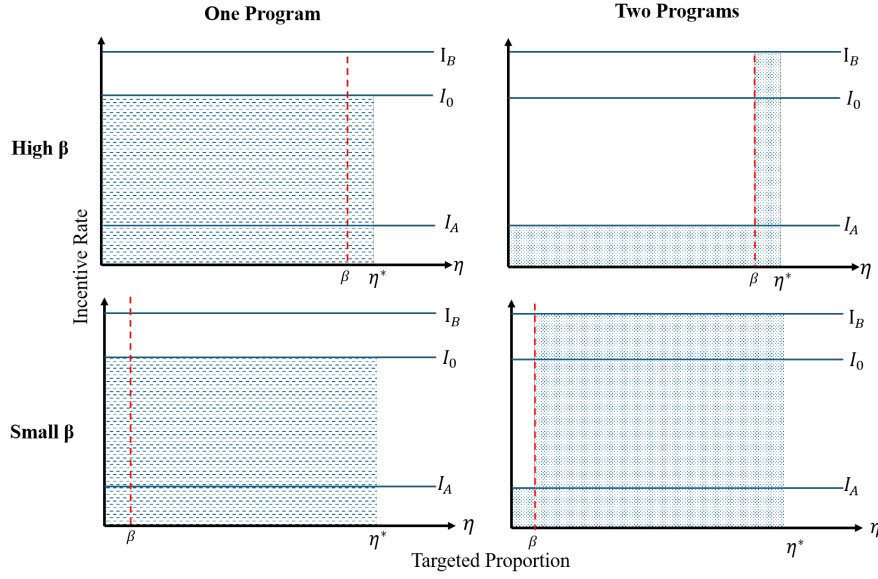


Figure 4: Case two, high target acres with few and many low-cost types

We compare the one program cost function from equation 21, $TC_1 = \eta * I_0$ and the two program cost function from equation 25, $TC_2 = \beta\eta I_A + (1 - \beta)\eta I_B$.

$$TC_2 \geq TC_1$$

$$\beta\eta I_A + (1 - \beta)\eta I_B \geq \eta * I_0 \quad (32)$$

$$\beta I_A + (1 - \beta) I_B \geq I_0$$

Given that $I_A < I_0 < I_B$

$$\text{At } \beta = 0 : \quad TC_2 = I_B > I_0 = TC_1, \quad (33)$$

$$\text{At } \beta = 1 : \quad TC_1 = I_A < I_0 = TC_1.$$

Consider the two-program cost function, TC_2 :

$$TC_2 = \beta\eta I_A + (1 - \beta)\eta I_B$$

$$\frac{\partial TC_2}{\partial \beta} = \eta(I_A - I_B) < 0 \quad (34)$$

Holding the incentives constant, when $\beta = 0$, $TC_2 = \eta I_B$ and when $\beta = 1$, $TC_2 = \eta I_A$. However, TC_1 does not depend on the β . So as β increases from 0 to 1, TC_2 moves down, crossing TC_1 exactly once. Figure 5 shows the cost curves over the different levels of β .

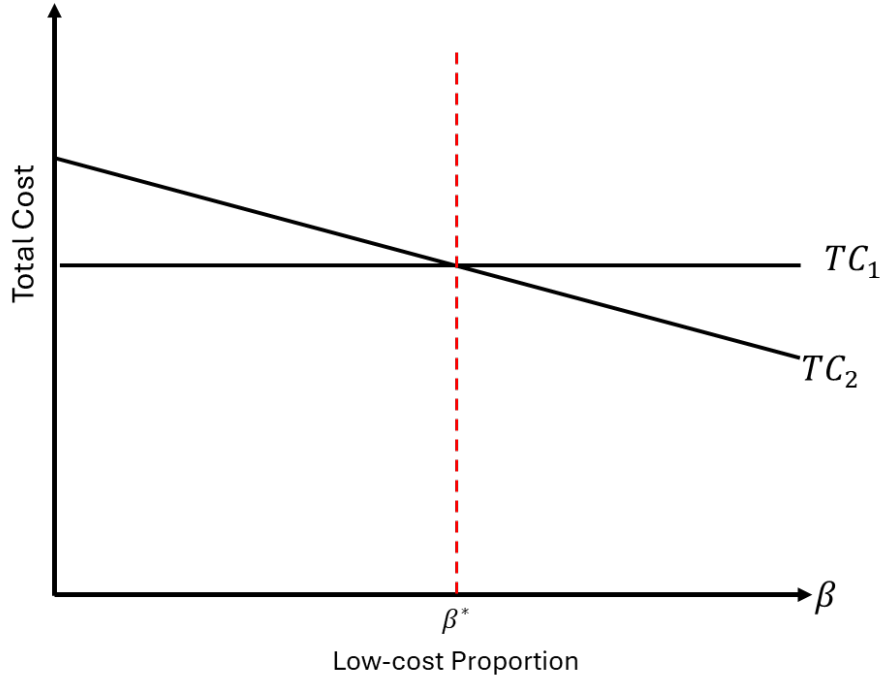


Figure 5: Cost functions over β values when $\eta^* > \beta$

5. Empirical Analysis

The theoretical model examines how producers self-select into conservation programs based on their underlying types and the cost implications of using one or two programs to meet a given acreage target. These types, denoted by θ , capture heterogeneity in production conditions such as soil characteristics, climate, and other field-specific attributes that are privately known to the producer. Within this framework, producers sort into different programs based on the relative returns from alternative land uses and the incentives offered.

In this section, we test whether characteristics that approximate θ , such as soil properties, climatic conditions, land capability classifications, and other factors, significantly influence producers' enrollment decisions, since θ is not directly observed but is approximated using these measurable variables.

Specifically, we model the probability that a field is enrolled in either CREP or EQIP as a function of climate, field characteristics, incentives, and other factors. Fields with more productive soils or favorable climate conditions may have higher returns under continued production or certain post-retirement uses, increasing their likelihood of enrolling in EQIP, which allows non-irrigated crop production after retirement.

5.1 Data

The empirical analysis integrates spatial, hydrological, soil, climatic, and agricultural datasets to construct a field-level dataset of land retirement in the Republican River Basin of Colorado. We obtained data on 429 fields participating in CREP and EQIP as of August 2025 from RRWCD. The data includes the field's spatial coordinates, field size, and the year each field was retired. We downloaded the shapefiles of irrigated fields from the Colorado Decision Support System (CDSS), which provides field boundaries. This shapefile identifies the wells used to irrigate each field. Combining these two datasets created a list of 3482 fields, both retired and irrigated, that we used in the analysis.

We derived non-irrigated land capability classifications and soil types from the USDA's Gridded Soil Survey Geographic Database (gSSURGO). The non-irrigated land capability class variable classifies land according to its suitability for agricultural production under non-irrigated conditions. Next, we intersected field boundaries with the land capability map unit polygons. Where fields overlap multiple map units, we calculated an area-weighted class. We expect fields that are considered not suitable for non-irrigated crop production with higher land classes to be more likely to join CREP, where they do not grow non-irrigated crops during the contract period

Measures of aquifer saturated thickness were taken from the 2009 High Plains aquifer dataset released by the USGS. We also downloaded the 2016 water table elevation from the USGS. The ground surface elevation at each well was retrieved from the USGS Elevation Point Query Service (EPQS), enabling us to calculate the depth to groundwater as the difference between the ground elevation and the water table elevation.

These climatic indicators are crucial for understanding the agro-ecological potential of retired fields. Higher summer temperatures or lower precipitation can increase the difficulty of maintaining non-irrigated crops and may alter the relative attractiveness of retirement compared to continuous irrigation. We obtained temperature and precipitation data from PRISM at 800-meter spatial resolution for the period 2005–2025. We calculated annual precipitation totals by summing monthly precipitation values within each calendar year. We calculated the mean summer temperature by first subsetting the monthly mean temperature data to the summer months (June, July, and August), then averaging these three months within each year, yielding a single mean summer temperature per year. We then averaged annual precipitation totals and mean summer temperatures to obtain a single mean value for each field.

Producers participating in land retirement programs receive financial incentives from the U.S. Department of Agriculture (USDA) and the Republican River Water Conservation District (RRWCD). These rates reflect per-acre payments offered under federal conservation programs such as CREP and EQIP. The USDA payment rates vary by county and year, as shown in table 1 and 2. We summed the 2024 RRWCD and USDA rates to determine the total CREP and EQIP rates each producer faces. The incentive rates are exogenously determined by policymakers and vary by county. This exogeneity implies that observed participation decisions primarily reflect producers’ opportunity costs and the program’s attractiveness.

5.2 Producer’s enrollment decision

The theoretical model developed in Section 3 characterizes producers as profit-maximizing agents who sort into conservation programs, or continue irrigating, based on an unobserved

type θ that governs their opportunity cost of retiring irrigated land. Producers with low θ , and therefore a low opportunity cost of foregoing irrigation, are predicted to sort into the program offering a lower incentive but permitting continued non-irrigated use, while producers with high θ require the larger, unconditional incentive of a stricter retirement program to be induced to enroll.

In this section, we then model the producer's choice among three alternatives, continued irrigation, enrollment in CREP, or enrollment in EQIP, using a random-utility framework. This specification allows us to empirically test whether the sorting behavior predicted by the theoretical model responds systematically to variation in opportunity cost. Similar to program A, EQIP offers lower incentives and allows non-irrigated crop production during and after the contract period. CREP is also similar to program B, which offers a higher incentive rate and prohibits the production of non-irrigated crops during the contract period.

For the producer i in a year t , the choice set is $\mathcal{K} = \{\text{Irrigate, CREP, EQIP}\}$. Utilities follow a random-utility structure:

$$U_{it} = (X_{it}\beta_k) + \varepsilon_{it} \quad (35)$$

Assume that ε_{it} are i.i.d. Type-I Extreme Value, then the choice probabilities are

$$Pr(Y_{it} = k) = \frac{\exp(X_{it}\beta_k)}{1 + \sum_{j=1}^K \exp(X_{it}\beta_j)} \quad (36)$$

We adopt a static model because the data are predominantly cross-sectional. CREP and EQIP incentive levels exhibit little temporal variation, and most heterogeneity is attributable to spatial differences across producers/fields rather than to changes over time. Thus, treating continued irrigation as the reference category, we can estimate the probability of producers to enroll in CREP, $Pr(Y_i = 1)$, and enroll in EQIP, $Pr(Y_i = 2)$, using the following probability functions:

$$\begin{aligned}
\Pr(Y_i = 0) &= \frac{1}{1 + \exp(X_i\beta_1) + \exp(X_i\beta_2)}, \\
\Pr(Y_i = 1) &= \frac{\exp(X_i\beta_1)}{1 + \exp(X_i\beta_1) + \exp(X_i\beta_2)}, \\
\Pr(Y_i = 2) &= \frac{\exp(X_i\beta_2)}{1 + \exp(X_i\beta_1) + \exp(X_i\beta_2)}.
\end{aligned} \tag{37}$$

In this study, we used data on 3,482 fields covering 478,813 acres that are CREP, EQIP, or irrigated production.² Table 3 shows the number of fields and acres used. There are 429 fields retired through CREP and EQIP covering about 57,445 acres, representing roughly 12% of the total acreage. Within this retired portion, CREP enrollments account for 210 fields and 30,909 acres (6.5% of total acres). The remaining 219 retired fields (26,536 acres, 5.5% of total acres) are enrolled in EQIP. The majority of the fields are in irrigated agricultural production: 3,053 fields covering 421,368 acres, representing 88.0% of total acres.

Table 3: Land use summary for fields used in the study

Land Use	Number of Fields	Acres	Percentage Acres
CREP	210	30909	6.5
EQIP	219	26536	5.5
Irrigating	3053	421368	88.0
Total	3482	478813	100

We include variables that explain the amount of groundwater available to the producer for irrigation, such as depth to groundwater and saturated thickness. We also include variables describing the soil characteristics, climatic conditions, and non-irrigated land class of the field. Table 4 summarizes the descriptive statistics of the key soil, climatic, and hydrological variables used in the analysis.

²The 62,396-acre basin-wide total reflects all CREP and EQIP retirements recorded by RRWCD since 2005. The empirical analysis in Section 4 uses a smaller sample of 429 fields covering 57,445 acres (30,909 acres CREP, 26,536 acres EQIP; see Table 3), because a subset of retired wells could not be matched to complete field-level records. Fields lacking complete data on spatial boundaries, soil characteristics, or climatic variables were excluded from the analysis sample.

Table 4: Descriptive Statistics for variables used

Variable	Units	Mean	Std Dev	Min	Max
Average Precipitation	Millimeters	488.30	18.33	416.29	525.27
Average Temperature	Degrees Celsius	22.55	0.26	21.25	23.50
CREP Rate	USD	3755.75	985.93	2310.00	6237.50
Depth to Groundwater	Meters	50.27	17.47	6.41	90.42
EQIP Rate	USD	1685.53	1371.21	994.45	4405.45
Field Size	Acres	137.08	59.56	4.70	880.00
Irrigated Class		2.87	0.81	1.00	6.00
Non Irrigated Class		3.61	1.14	2.00	7.00
Saturated Thickness	Feet	122.64	72.60	0.72	302.81

Average precipitation across the study fields is 488.30 mm, with relatively low variability ($SD = 18.33$ mm) and a range spanning 416.29 mm to 525.27 mm, indicating that fields in the basin experience fairly similar precipitation conditions. Average temperature is likewise tightly distributed, with a mean of 22.55°C and a standard deviation of only 0.26°C, ranging from 21.25°C to 23.50°C. Depth to groundwater shows considerably greater spatial heterogeneity, ranging from 6.41 m to 90.42 m with a mean of 50.27 m, reflecting substantial variation in aquifer accessibility across fields.

Similarly, saturated thickness averages 122.64 ft but ranges widely from 0.72 ft to 302.81 ft. Field size averages 137.08 acres ($SD = 59.56$), with a maximum of 880.00 acres, while the mean non-irrigated land capability class of 3.61 indicates that, absent irrigation, most fields face moderate limitations on crop production, and is higher than the mean irrigated class of 2.87. The incentive variables show substantial dispersion: CREP rates average \$3,755.75 per acre ($SD = 985.93$, range \$2,310.00–\$6,237.50), while EQIP rates average \$1,685.53 per acre but exhibit even greater relative variability ($SD = 1,371.21$, range \$994.45–\$4,405.45).

The distribution of low- and high-opportunity cost fields, together with the size of the acreage-retirement target, determines whether a single program or a mix of programs offers the lowest-cost path to meeting conservation goals, as established in Section 3.2. We can use the land capability classes as a proxy for θ . Figure 6 shows the distribution of fields across non-irrigated land capability classes. Higher non-irrigated classes correspond to land that faces more severe limitations and therefore yields lower returns under non-

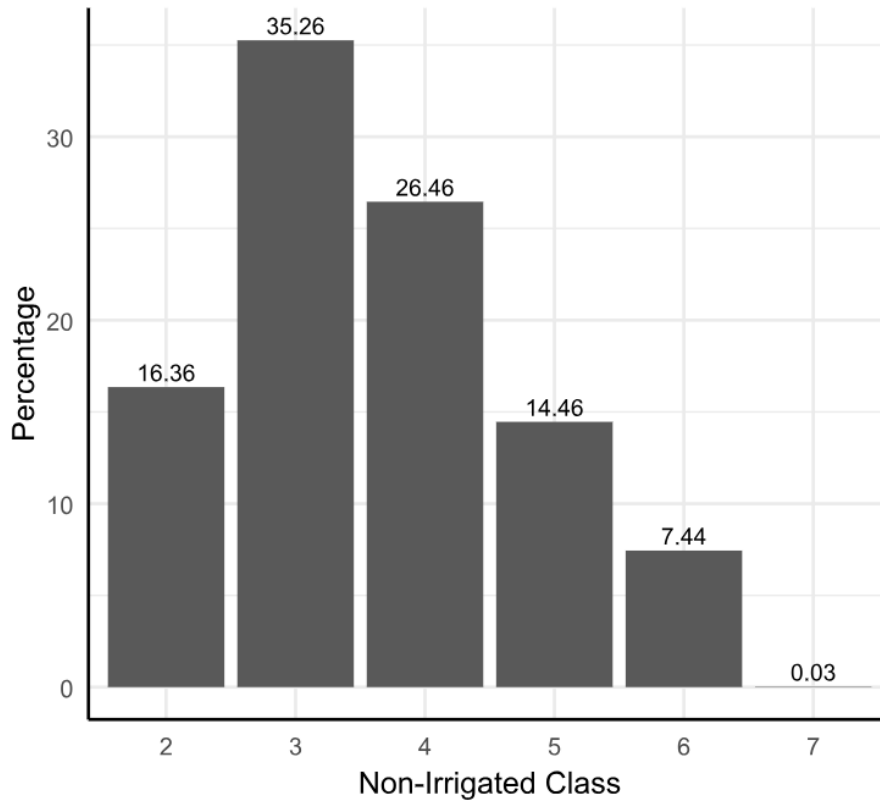


Figure 6: Percentage of fields in each non-irrigated land class

irrigated production.

The distribution is concentrated in class 3, which accounts for 35.26% of fields, followed by class 4 at 26.46%. Only 16.36% of fields fall in class 2, the lowest class represented, while classes 5, 6, and 7 together account for roughly 22% of fields, with class 7 fields being negligible (0.03%). Hence, the majority of fields, over 83%, fall in class 3 or above, indicating that most land in the study area faces moderate to severe limitations on non-irrigated crop production.

In terms of the theoretical framework in Section 3, this suggests that the share of low-opportunity-cost fields (β), which are better suited for non-irrigated use and thus more likely to enroll in program A, is relatively small. Considering the total acres retired since 2005 (62,396 acres or 13.5%), and that the programs are still enrolling, this distribution is consistent with the case-two scenario described in Section 3.2.2, where the retirement target is likely to exceed the pool of available low-cost fields, implying that a mixed-program design combining CREP and EQIP, rather than a single uniform-incentive program, may be the more cost-effective approach to reaching basin-wide acreage-retirement goals.

We calculated the pairwise correlation coefficients among the variables shown in table 5. A very high correlation appears between the two incentive variables, CREP Rate and EQIP Rate, with a coefficient of 0.91. A moderate correlation is observed between precipitation change and saturated thickness ($r = 0.40$), reflecting shared hydrogeological conditions across areas. Sand soil is correlated with the irrigated land class, non-irrigated land class, and saturated thickness ($r = -0.50$, $r = 0.56$, and $r = 0.43$, respectively).

Table 5: Correlation matrix

	Average Precipitation	Average Temperature	CREP Rate	Depth to Groundwater	EQIP Rate	Field Size	Irrigated Class	Non Irrigated Class	Sandy Soil	Saturated Thickness
Average Precipitation	1.00	0.15	0.29	-0.07	0.33	-0.02	-0.03	-0.06	-0.06	0.05
Average Temperature	0.15	1.00	0.20	-0.12	0.18	0.02	-0.08	-0.07	-0.08	-0.02
CREP Rate	0.29	0.20	1.00	-0.06	0.91	-0.13	0.06	0.03	-0.10	-0.20
Depth to Groundwater	-0.07	-0.12	-0.06	1.00	0.05	0.08	-0.39	-0.43	-0.30	-0.24
EQIP Rate	0.33	0.18	0.91	0.05	1.00	-0.13	-0.04	-0.09	-0.25	-0.43
Field Size	-0.02	0.02	-0.13	0.08	-0.13	1.00	0.02	0.05	0.07	0.11
Irrigated Class	-0.03	-0.08	0.06	-0.39	-0.04	0.02	1.00	0.82	0.50	0.27
Non Irrigated Class	-0.06	-0.07	0.03	-0.43	-0.09	0.05	0.82	1.00	0.56	0.35
Sandy Soil	-0.06	-0.08	-0.10	-0.30	-0.25	0.07	0.50	0.56	1.00	0.43
Saturated Thickness	0.05	-0.02	-0.20	-0.24	-0.43	0.11	0.27	0.35	0.43	1.00

Finally, the irrigated land class variable shows a strong positive correlation with the Non-Irrigated land class ($r = 0.82$), indicating that these land-use classifications move together. Overall, the correlations highlight soil type as a key factor associated with both irrigated and non-irrigated land classes.

5.3 Enrollment Decision Results

Table 6 presents the estimated coefficients from the MNL model of program participation, with continued irrigation as the reference category. Based on these correlations, we exclude EQIP Rate, Irrigated Class, and Sandy Soil from the final specification reported in Table 6. Including both would introduce substantial multicollinearity.³

³To address the correlation between CREP and EQIP incentive rates, we also explored an orthogonalization (residualization) strategy to isolate EQIP payment variation independent of CREP incentives.

Table 6: Multinomial logit Results on choice of conservation program.

Variable	Estimate	Std. Error	z value	Pr(> z)
CREP				
Intercept	-8.4437	6.1547	-1.3719	0.1701
Average Precipitation	-0.1772	0.1160	-1.5280	0.1265
Average Temperature	0.2623	0.2680	0.9787	0.3277
CREP Rate	0.0060	0.0008	7.9548	0.0000
Depth to Groundwater	0.0017	0.0046	0.3657	0.7146
Field Size	0.0043	0.0010	4.1684	0.0000
Non Irrigated Class	0.4141	0.0739	5.6023	0.0000
Saturated Thickness	-0.0135	0.0016	-8.6155	0.0000
EQIP				
Intercept	-11.9227	5.2880	-2.2547	0.0242
Average Precipitation	0.0546	0.1090	0.5013	0.6162
Average Temperature	0.3712	0.2301	1.6131	0.1067
CREP Rate	0.0059	0.0006	9.2006	0.0000
Depth to Groundwater	0.0000	0.0040	0.0082	0.9935
Field Size	-0.0025	0.0015	-1.6962	0.0899
Non Irrigated Class	-0.0896	0.0761	-1.1782	0.2387
Saturated Thickness	-0.0193	0.0020	-9.5927	0.0000
Reference Category = Irrigation				
Residual deviance: 2824.15, Log-likelihood: -1412.075, n = 3534				

Three observations were dropped due to missing values.

CREP enrollment is significantly associated with the financial incentives and land characteristics. The CREP payment rate is positive and highly significant (Estimate = 0.006, $p < 0.0008$), indicating that higher incentive payments raise the probability of CREP enrollment. Field size has a small but statistically significant positive effect (Estimate = 0.0043, $p = 0.000$), indicating that larger fields are more likely to be enrolled in CREP.

The non-irrigated land class coefficient is positive and highly significant (Estimate = 0.4121, $p < 0.0001$), indicating that fields facing greater limitations under non-irrigated production, and therefore a higher opportunity cost of irrigation, are more likely to enroll in CREP. This is consistent with the prediction that producers with higher opportunity costs are more likely to enroll in a high-paying, more restrictive program such as CREP, in which they do not have to engage in non-irrigated production. Saturated thickness is negative and highly significant (Estimate = -0.0135, $p < 0.0001$), implying that fields with greater aquifer access, and correspondingly higher expected returns to continued

EQIP rates were regressed on CREP rates, and the resulting residuals were used as an alternative to EQIP rates in the model. However, the results remained the same, and the EQIP residuals were insignificant.

irrigation, are less likely to enroll in CREP. Depth to groundwater, precipitation, and temperature are not statistically significant predictors of CREP enrollment.

EQIP enrollment follows a broadly similar pattern with some notable differences. The CREP rate variable is also positive and significant in the EQIP equation (Estimate = 0.0059, $p < 0.0001$). These results, together with the high correlation observed in the correlation analysis, suggest that higher incentives increase enrollment in EQIP.

In contrast to CREP, the non-irrigated class coefficient is negative and not significant for EQIP (Estimate = -0.0866, $p < 0.2387$). Although the effect is insignificant, the negative sign indicates that fields in higher non-irrigated land classes are less likely to enroll in EQIP. This divergence between the two programs makes intuitive sense given their contract structures. EQIP permits non-irrigated crop production after retirement, so fields in low non-irrigated land class (land capable to support non irrigated crop production), and correspondingly lower opportunity cost of irrigation, are more attracted to it relative to CREP. This is consistent with the theoretical prediction that producers with lower opportunity costs enroll in a less restrictive program, such as EQIP. Finally, saturated thickness is negative and significant for EQIP (Estimate = -0.0193, $p < 0.0001$). It has the same effect as on CREP enrollment: fields with greater access to irrigation water are less likely to retire from irrigated production.

Taken together, these results indicate that while financial incentives and aquifer conditions influence enrollment in both CREP and EQIP in similar directions, the two programs diverge sharply in how they respond to non-irrigated land suitability and field size, consistent with the theoretical prediction that producers self-select into the program that best matches their underlying opportunity cost and the post-retirement land-use restrictions each program imposes.

6. Conclusion

This study examines participation in conservation programs to retire irrigated land and develops a theoretical framework for understanding when a single conservation program

versus a mix of programs delivers the lowest-cost path to a given acreage-retirement target. The theoretical model shows that cost-effectiveness depends critically on the size of the retirement target relative to the supply of low-opportunity-cost producers. When the target is small and can be met entirely by low-cost types, a single program offering a uniform incentive is the least-cost option. As the target grows and begins to require enrollment of higher-cost producers, a two-program design that allows self-selection, pairing a lower-incentive program permitting continued non-irrigated use with a higher-incentive program for stricter retirement, becomes more cost-effective.

The empirical analysis found that program participation is shaped by a combination of financial incentives and underlying field characteristics. Higher payment rates significantly increase the likelihood of enrollment in both programs relative to continued irrigation, consistent with the theoretical framework. Most notably, the two programs diverge in how they respond to non-irrigated land class and field size. Fields in higher non-irrigated land capability classes, meaning land poorly suited to non-irrigated production and thus facing a higher opportunity cost of irrigation, are more likely to enroll in CREP.

This study has several limitations. First, the analysis relies on a static, cross-sectional framework that does not capture how enrollment decisions or incentive levels may evolve as the program matures or groundwater conditions change over time. Second, producer type (θ) is only indirectly proxied through observable non-irrigated land class, and climatic variables, and may not fully capture privately known information that drives actual enrollment decisions, such as risk preferences, farm financial condition, or off-farm income. Third, the data are specific to the Republican River Basin, and generalizing the findings to other basins or conservation contexts with different program designs and producer populations remains an open question. Future research could extend this framework using panel data to examine dynamic enrollment and re-enrollment behavior, or could incorporate richer producer-level data to better capture the heterogeneity underlying program self-selection.

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